

# TEJAS PATTABHI

Staff Engineer — Cloud · Distributed Systems · Applied AI

San Diego, CA · [tejas.pattabhi@gmail.com](mailto:tejas.pattabhi@gmail.com) · +1 (469) 570-1988 · [tejaspattabhi.com](http://tejaspattabhi.com) · [linkedin.com/in/tejas-pattabhi](https://linkedin.com/in/tejas-pattabhi)

## SUMMARY

Staff Engineer with 10+ years turning large-scale data and compute problems into reliable production systems. At Qualcomm, works on the critical path of the autonomous-driving program — the infrastructure and applied AI that validate the software running in production vehicles. Named co-inventor on two granted U.S. patents.

## CORE SKILLS

|                         |                                                            |                   |                                                                                                               |
|-------------------------|------------------------------------------------------------|-------------------|---------------------------------------------------------------------------------------------------------------|
| <b>Languages</b>        | Python · Java · C# · C · Bash                              | <b>Cloud</b>      | AWS · Google Cloud (Dataflow) · Azure                                                                         |
| <b>Orchestration</b>    | Kubernetes · Argo Workflows/Events · Docker · AWS CDK      | <b>Applied AI</b> | LLM agents (Claude Agent SDK · Bedrock) · agentic coding (Claude Code · Codex) · custom Claude skills/plugins |
| <b>ML &amp; vectors</b> | Self-hosted embeddings (Qwen3) · pgvector · semantic dedup |                   |                                                                                                               |

## EXPERIENCE

### Qualcomm

Sep 2021 – Present

#### Staff Engineer · San Diego, CA

*Build the large-scale infrastructure that replays recorded drive data to validate automated-driving software — perception, radar, vehicle self-motion/positioning, ground-truth labeling, and automatic emergency braking.*

- **Automotive-grade reprocessing clusters** — designed, built, and operate two on-prem Kubernetes clusters of Qualcomm **QCR-100** cards (the same Snapdragon Ride silicon that ships in vehicles) for bit-accurate ADAS/AD reprocessing, running **~100–300x faster than general-purpose compute** — fast enough to replay recorded drives at close to real time.
- **Reprocessing-as-a-Service platform** running identically in cloud or on-prem — Kubernetes job orchestration, DAG dependency resolution, event-driven **Argo Workflows**, parallel petabyte-scale data transfer, and automated failure classification. Also integrated the QCR-100 into custom hosted cloud environments for on-demand partner/customer access.
- **Key collaborator on the Qualcomm–BMW Group program** — helped deliver the reprocessing pipeline behind the jointly developed Snapdragon Ride Pilot system (publicly launched Sept. 2025, debuting in the BMW iX3).
- **Production AI triage agent** (project lead) handling tens of thousands of failures/day — self-hosted embeddings + **pgvector** semantic dedup, then a **Claude Agent SDK + AWS Bedrock** agent that diagnoses root cause and opens Jira tickets / GitHub fix-PRs, escalating ambiguous cases to humans.
- **Fleet-health & accuracy monitoring** — Prometheus stack tracking each accelerator's health/firmware and **Real-Time Factor (RTF)**, the measure of how closely reprocessing matches ground truth; plus IPMI/BMC power control and safe Kubernetes node lifecycle for servicing without disrupting jobs.

### OpenText (formerly Webroot)

Sep 2015 – Sep 2021

#### Senior Software Engineer · San Diego, CA

*Led internet-scale cybersecurity and threat-intelligence systems.*

- Led distributed clusters that safely crawled and threat-classified a **peak of 17M+ URLs/day**; co-invented "DeepCrawl," which discovers related malicious resources from a known link and classifies threat level with an ML model — **basis of two granted U.S. patents**.
- Built an IP threat-classification system and a cloud-reputation scoring system, and architected multi-cloud honeypots (AWS, GCP, Azure) capturing live attacks to feed both. Owned AWS hosting/DevOps.

**Earlier:** Award Solutions — UT Dallas OpenStack project, open-sourced (2015) · TradeStation Technologies — SDE Intern, AWS EMR log analytics (2014) · Société Générale — Windows Server Administrator, Bangalore (2012–2013).

## PATENTS · PUBLICATIONS · EDUCATION

**U.S. Patents 12,452,259 B2 (2025) & 11,201,875 B2 (2021)** — "Web Threat Investigation Using Advanced Web Crawling," named co-inventor, Open Text Inc. · First-author publication, *IJITCS*, 2013 (spatial-clustering algorithm).

**M.S. Computer Science** — UT Dallas · 2015

**B.E. Information Science** — PES / VTU, India · 2012 · First Class with Distinction (all 8 semesters)